

Module Review: L-carnitine dehydrogenation (UniPathway UPA00117) in *Pseudomonas putida* KT2440

Target: PSEPK / *P. putida* KT2440 (NCBI taxon 160488, proteome UP000000556) **Bucket:** unipathway:UPA00117 — NAD(+)-dependent oxidation of L-carnitine to 3-dehydrocarnitine
Candidate gene: lcdH / PP_0302 / Q88R32 (L-CDH, EC 1.1.1.108)

1. Executive summary

The scoped UPA00117 step — NAD(+)-dependent oxidation of L-carnitine to 3-dehydrocarnitine (Rhea RHEA:19265, EC 1.1.1.108) — is **COVERED** in *P. putida* KT2440 by the single candidate gene **lcdH/PP_0302 (Q88R32)**. Confidence is **high**, supported by three converging lines of evidence: (i) a specific KEGG orthology assignment K17735 "carnitine 3-dehydrogenase" (not the generic fatty-acid β -oxidation KO); (ii) membership in the L-carnitine dehydrogenase subfamily of the 3-hydroxyacyl-CoA dehydrogenase family; and (iii) location within a dedicated carnitine-catabolism gene cluster (transporter + regulator + downstream cleavage enzymes) that mirrors the experimentally validated *P. aeruginosa* carnitine operon.

The main caveat is that there is **no direct biochemical or genetic characterization of PP_0302 itself** — the UniProt entry is protein-existence level 3 ("Inferred from homology"), with the genome paper as its only reference. However, direct enzymology exists for the carnitine dehydrogenase of *P. putida* strain IFP 206 (same species), giving strong same-species transfer support. No module revision is needed; the single-step bucket is satisfiable and correctly bounded.

2. Target-organism pathway definition

- **Included (scoped UPA00117):** the *single* enzymatic reaction L-carnitine + NAD(+) $\rightarrow$ 3-dehydrocarnitine + NADH + H(+). This is a carnitine-*catabolic* (degradative) dehydrogenase

step, not carnitine biosynthesis and not the CoA-dependent CaiB/CaiD carnitine racemase/dehydratase (anaerobic *Enterobacteriaceae*) route.

- **Kept separate (neighboring pathways / broad maps):**
- Downstream carnitine catabolism: 3-dehydrocarnitine → glycine betaine → dimethylglycine → sarcosine → glycine (a distinct, larger pathway; genomically present in KT2440 but outside this bucket).
- Glycine betaine biosynthesis/osmoprotection (choline → GB) — shares the GB metabolite node but is a different process.
- Fatty-acid β -oxidation (EC 1.1.1.35, 3-hydroxyacyl-CoA dehydrogenase / FadB) — same protein *superfamily*, different substrate; a classic over-propagation trap.
- Anaerobic carnitine metabolism (CaiTABCDE, crotonobetaine/ γ -butyrobetaine) — not this pathway.
- **Alternate names / DB definitions:** L-carnitine dehydrogenase = carnitine 3-dehydrogenase = carnitine:NAD⁺ oxidoreductase; EC 1.1.1.108; Rhea RHEA:19265; KO K17735; UniProt PATHWAY "Amine and polyamine metabolism; carnitine metabolism"; gene symbol *lcdH* (also *cdh* in some *Pseudomonas* literature).

3. Expected step model

UPA00117 as seeded is a **one-step module**:

Step	Reaction	Expected enzyme	Status in KT2440
S1. L-carnitine dehydrogenation	L-carnitine + NAD ⁺ → 3-dehydrocarnitine + NADH + H ⁺	Carnitine 3-dehydrogenase (EC 1.1.1.108, K17735)	Covered by PP_0302

There are no additional expected internal steps within this bucket ("No explicit connections"). The provisional outline is accurate: the module is the dehydrogenase step only, not the full utilization route.

4. Candidate genes and evidence

lcdH / PP_0302 / Q88R32 — L-carnitine dehydrogenase (EC 1.1.1.108)

- **Likely role:** genuine carnitine 3-dehydrogenase catalyzing step S1. High confidence.
- **Evidence type & strength:**
- *Sequence/annotation:* 321 aa, 34.6 kDa/subunit, annotated homodimer (~69 kDa). Matches the biochemically characterized *P. putida* IFP 206 enzyme (homodimer, ~62 kDa, NAD⁺- and L-carnitine-specific, EC 1.1.1.108; Goulas 1988,

P 3058208

— **strong same-species transfer.**

- *Ortholog specificity:* KO **K17735** (carnitine 3-dehydrogenase) is substrate-specific, distinguishing PP_0302 from generic 3-hydroxyacyl-CoA dehydrogenases (fadB/K01825). Family: 3-hydroxyacyl-CoA dehydrogenase family, **L-carnitine dehydrogenase subfamily** (Pfam PF02737 NAD-binding + PF00725 C-terminal; COG1250; PANTHER PTHR48075:SF5).
- *Genomic context (strong):* PP_0302 lies in a carnitine-catabolism cluster — PP_0301 betainyl-CoA thiolase, PP_0303 dehydrocarnitine cleavage enzyme, PP_0304 carnitine ABC-transporter periplasmic component, PP_0305 regulator CdhR, PP_0298 AraC-family regulator. This synteny parallels the transposon-validated *P. aeruginosa* carnitine operon (PA5388–PA5384 + PA1999–PA2000; Wargo & Hogan 2009,

P 19406895

- **Curation-relevant caveats:**
- UniProt protein existence **level 3 (inferred from homology)**; no direct KT2440 assay or knockout — function is *predicted*, not *demonstrated*, for this specific ORF.
- EC 1.1.1.108 and the Pfam domains belong to a broad NAD-dependent dehydrogenase superfamily; the confident call depends on the specific KO + subfamily + operon context, not on Pfam alone.
- Rhea/EC mapping is exact and appropriate; no over-broad GO expected here (GO:0008859 carnitine 3-dehydrogenase activity is specific and correct).

5. Gaps, ambiguities, and likely over-annotations

- **No gap within the bucket:** S1 is covered; the module is satisfiable.

- **Paralog ambiguity — RESOLVED:** a KEGG ortholog check confirms PP_0302 is the **sole** carnitine-3-dehydrogenase (K17735) gene in KT2440; the generic 3-hydroxyacyl-CoA dehydrogenase KO K00022 has 0 KT2440 genes, and the fatty-acid β -oxidation *fadB* KO K01825 maps to a **separate** locus (PP_2136, not PP_0302). Thus there is no competing carnitine-dehydrogenase paralog, and the superfamily's β -oxidation activity is carried by a distinct gene. The step assignment to PP_0302 is unambiguous at the ortholog level.
- **Over-annotation risk elsewhere, not here:** the reverse risk (a true *fadB* being called *lcdH*) is the more common error in this family; PP_0302's cluster context argues against it being a mis-annotation.
- **Scope creep risk:** the downstream carnitine→glycine betaine→glycine genes (PP_0301, PP_0303, transporter PP_0304, regulator PP_0305) are present but must **not** be folded into UPA00117; they belong to separate downstream modules. Keeping the bucket single-step is correct.
- **Physiology unverified:** whether KT2440 actually grows on L-carnitine as C/N source is not directly documented here (demonstrated for *P. aeruginosa*, and the complete cluster in KT2440 makes it likely). This is an open question, not a module defect.

6. Module and GO-curation recommendations

- **Step S1 (L-carnitine dehydrogenation):** mark **covered** via *lcdH*/PP_0302. Evidence code $\approx$ inferred by sequence/orthology + genomic context (ISS/IBA-type), *not* experimental for this ORF.
- **Module boundaries:** correct as scoped; **no** **module_needs_revision**. Do not expand UPA00117 to include downstream GB-forming steps.
- **GO annotation:** GO:0008859 (carnitine 3-dehydrogenase activity) and GO:0009437 (carnitine metabolic process) for PP_0302 are appropriate and specific; no new GO term request needed.
- **No** **gap**, **candidate_uncertain**, or **not_expected_in_target_taxon** flags required for the single scoped step. If a broader carnitine-utilization module is later created, PP_0301/PP_0303/PP_0304/PP_0305 would populate its downstream/transport/regulatory steps.

7. Genes to promote to full review

- **PP_0302 / lcdH (Q88R32):** promote to full `fetch-gene` review to (a) record that it is confirmed as the **sole K17735 copy** in KT2440 (fadB/ β -oxidation is the separate PP_2136), (b) attach the same-species biochemical reference (IFP 206) and the *P. aeruginosa* operon precedent, and (c) upgrade the evidence narrative from bare homology to homology + unique-ortholog + synteny. This is the only gene in the bucket and warrants a curated record.
- (Context only, not in this bucket) PP_0301, PP_0303, PP_0304, PP_0305 merit review when the downstream carnitine-catabolism / GB module is curated.

8. Key references

- Goulas P. (1988) *Purification and properties of carnitine dehydrogenase from Pseudomonas putida*. PMID **3058208**. — Direct same-species biochemistry: EC 1.1.1.108, L-carnitine $\rightarrow$ 3-dehydrocarnitine, NAD⁺-specific, homodimer.
 - Wargo MJ, Hogan DA. (2009) *Identification of genes required for Pseudomonas aeruginosa carnitine catabolism*. PMID **19406895**. — Genetic definition of a *Pseudomonas* carnitine operon (carnitine $\rightarrow$ 3-dhc $\rightarrow$ glycine betaine), supporting the KT2440 cluster interpretation.
 - Nelson KE et al. (2002) Complete genome sequence of *P. putida* KT2440. PMID **12534463**. — Source annotation of PP_0302/lcdH.
 - UniProtKB **Q88R32** (LCDH_PSEPK); KEGG **ppu:PP_0302 / KO K17735**; Rhea **RHEA:19265**; UniPathway **UPA00117**.
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Uncertainty ledger

- **Direct for KT2440:** genome annotation, gene-cluster synteny, KO/Rhea/EC mapping.
- **Same-species transfer (strong):** biochemistry from *P. putida* IFP 206.
- **Related-species transfer (moderate–strong):** operon logic and carnitine-catabolism physiology from *P. aeruginosa*.
- **Inferred/unverified:** actual enzymatic assay of PP_0302, KT2440 growth on carnitine. (Note: targeted PubMed searches returned no KT2440-specific carnitine growth, knockout, or transcriptomic study — the assignment therefore rests on homology + unique-ortholog +

operon synteny, with same-species biochemistry from *P. putida* IFP 206 and genetic validation from *P. aeruginosa* as the strongest transferable support.)

- **Resolved:** absence of confounding paralogs (PP_0302 is the sole K17735 gene; fadB is the separate PP_2136).

Curation decision (one line)

UPA00117 step **S1** = **covered** by lcdH/PP_0302; module boundaries correct (single-step); no new module or GO-term request needed; promote PP_0302 to full **fetch-gene** review. Sole residual gap = no direct KT2440 enzyme assay/knockout.

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